partial ordered amplitude

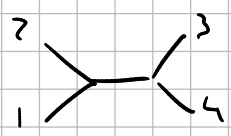
$\mathrm{Res}\, A_n(1\,2\,3 \ldots n)$ w.r.t one of the $X_{ij}$ (the poles)

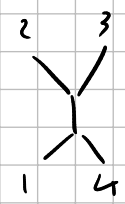
factorizes as product of two smaller amplitudes

if $f(z)$ is a complex function with Laurent series

$$= \sum_{n\in\mathbb{Z}} a_n z^n$$

$\mathrm{Res}\, f = a_{-1}$

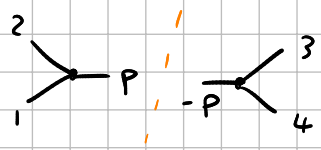



$$A_4(1234) \propto \frac{1}{s_{12}} + \frac{1}{s_{23}}$$

look at residue near one of the poles (ex: $s_{12}$)

$$\mathrm{Res}\,(A_4)_{s_{12}} = 1$$




$$\propto \int dp\; A_3(12p)\, A_3(-p\,34)$$

$$A_4 = \frac{1}{s_{12}} \boxed{A_{left}\; A_{right}} + O(1)$$

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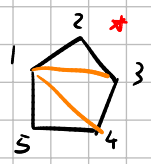
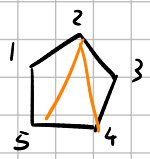
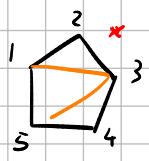
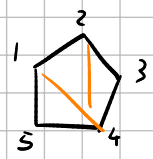
if we want to expand at $s_{12} = X_{13}$

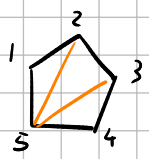



$$A_5(12345) = \boxed{\frac{1}{X_{13}X_{14}} + \frac{1}{X_{13}X_{35}}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{24}X_{14}} + \frac{1}{X_{25}X_{35}}$$



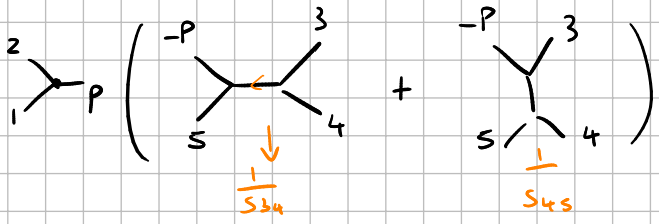




$$\mathrm{Res}\,(A_5)_{X_{13}} = \frac{1}{X_{14}} + \frac{1}{X_{35}} =$$

$s_{45}$ $\quad$ $s_{34}$

$(p_1+p_2+p_3)^2$

$= (p_4+p_5)^2$



$$= A_3(12p)\, A_4(-p345)$$

this follows from $S^\dagger S = \mathbb{1}$ (S-matrix)

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take amplitude $A_n$

choose 2 momenta $i, j$ $\qquad p_i \to p_i + zq \qquad z \in \mathbb{C}$

$\qquad\qquad\qquad\qquad\qquad\qquad p_j \to p_j - zq$

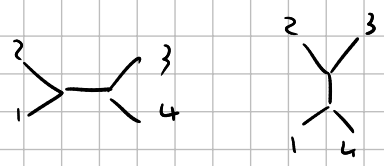
$q$ is a momentum vector s.t. $\quad p_i \cdot q = p_j \cdot q = q^2 = 0$

this does not effect that $p_i^2 = p_j^2 = 0$ and does not change $s_{ij}$

$$(p_i + zq)^2 = p_i^2 + z\underbrace{q \cdot p_i}_{=0} + z^2 \underbrace{q^2}_{=0} = p_i^2$$

the new amplitude depends on $z$

$$A(0) = \oint dz \, \frac{A(z)}{z}$$




<u>ex:</u> $\quad A_4(1234) = \dfrac{1}{s_{12}} + \dfrac{1}{s_{23}}$

$p_1 \to p_1 + qz \qquad\qquad s_{12} \to s_{12} = (p_1 + p_2)^2$

$p_2 \to p_2 - qz \qquad\qquad s_{23} \to (p_2 - qz + p_3)^2$

$$A_4(1234)(z) = \frac{1}{(p_1+p_2)^2} + \frac{1}{(p_2 - qz + p_3)^2}$$

$$\frac{A(z)}{z} = \frac{A(0)}{z} + A'(0) + \frac{A''(0)}{2} z + \dots$$

$A_4(0)$ = original amplitude

$\dfrac{A(z)}{z}$ pole at $z=0$ $\qquad$ expand in a series: $\quad \mathrm{Res}\, \dfrac{A(z)}{z} = A(0)$

$$A(0) = \mathrm{Res}\ \frac{A(z)}{z} \propto \oint \frac{A(z)}{z}\, dz$$

$$s_{14} = (P_1 + P_4)^2 =$$

look at $A_4(1234) = \frac{1}{s_{12}} + \frac{1}{s_{14}} = \frac{1}{s_{12}} - \frac{1}{s_{12} + s_{13}}$

$$s_{12} + s_{13} = (P_1 + P_2)^2 + (P_1 + P_3)^2 = 2P_1 \cdot P_2 + 2P_1 \cdot P_3 = 2\, P_1 \cdot (P_2 + P_3)$$

$$= 2\, P_1 (P_1 + P_2 + P_3) = 2\, P_1 \cdot (-P_4) = -(P_1 + P_4)^2 = -s_{14}$$

if $s_{13} = 0$ then $A_4(1234) = 0$

$$A_4 = \frac{1}{2\, P_1 \cdot P_2} + \frac{1}{2\, P_1 \cdot P_4}$$

$$P_1 \cdot P_3 = 0$$

$$P_1 \cdot (P_1 + P_2 + P_3 + P_4) = 0$$

$$\downarrow$$

$$P_1 \cdot P_2 = -P_1 \cdot P_4$$

$$A_5 = \frac{1}{x_{13}\, x_{14}} + \frac{1}{x_{13}\, x_{35}} + \frac{1}{x_{24}\, x_{25}} + \frac{1}{x_{24}\, x_{14}} + \frac{1}{x_{25}\, x_{35}}$$

$$= \frac{1}{s_{12}\, s_{45}} + \frac{1}{s_{23}\, s_{15}} + \frac{1}{s_{34}\, s_{21}} + \frac{1}{s_{45}\, s_{32}} + \frac{1}{s_{51}\, s_{43}}$$

$$P_1 + P_2 + P_3 + P_4 + P_5 = 0 \qquad P_i^2 = 0$$

$$s_{ij} = 2\, P_i \cdot P_j$$

$$\frac{1}{s_{12}} \left( \frac{1}{s_{45}} + \frac{1}{s_{34}} \right) + \frac{1}{s_{23}} \left( \frac{1}{s_{15}} + \frac{1}{s_{45}} \right) = -\frac{1}{s_{15}\, s_{43}}$$

$$\frac{1}{s_{12}}\left(\frac{s_{15}s_{43}}{s_{45}} + s_{15}\right) + \frac{1}{s_{23}}\left(s_{43} + \frac{s_{15}s_{43}}{s_{45}}\right) = -1$$

$s_{43} = 0$ $\qquad$ $\dfrac{s_{15}}{s_{12}} = -1$

$s_{13} = s_{14} = 0$

$$\frac{1}{s_{12}\,s_{45}} + \frac{1}{s_{23}\,s_{15}} + \frac{1}{s_{34}\,s_{21}} + \frac{1}{s_{45}\,s_{32}} + \frac{1}{s_{51}\,s_{43}}$$

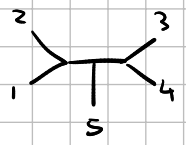
| | | |
|---|---|---|
| mny | 13 | 14 |
| mny | 24 | 25 |
| | 31 | 35 |
| | 42 | 41 |
| | 52 | 53 |

$p_1 \cdot p_3 = 0$ $\qquad$ $p_1 \cdot p_4 = 0$ $\qquad$ $p_1 \cdot p_2 = -p_1 \cdot p_5$

$p_1 \cdot (p_2 + p_3 + p_4 \cdot p_5) = 0$ $\qquad$ $s_{15} = -s_{12}$

$$\cancel{\frac{1}{s_{12}\,s_{45}}} - \frac{1}{s_{23}\,s_{12}} + \cancel{\frac{1}{s_{34}\,s_{12}}} + \frac{1}{s_{45}\,s_{32}} - \cancel{\frac{1}{s_{12}\,s_{43}}}$$

$$\frac{1}{s_{12}\,s_{34}}$$

$A_5 =$ 

$+$ cyclic

$$= \frac{1}{s_{12}\,s_{34}} + \frac{1}{s_{23}\,s_{45}} + \frac{1}{s_{34}\,s_{51}} + \frac{1}{s_{45}\,s_{12}} + \frac{1}{s_{51}\,s_{23}}$$

$p_1 \cdot p_3 = 0$ $\qquad$ $p_1 \cdot p_4 = 0$ $\qquad$ $p_1 \cdot p_2 = -p_1 \cdot p_5$

$p_1 \cdot (p_2 + p_3 + p_4 \cdot p_5) = 0$ $\qquad$ $s_{15} = -s_{12}$

$p_1 \cdot p_2 + p_1 \cdot p_5 = 0$

$$\cancel{\frac{1}{s_{12}\, s_{34}}} + \frac{1}{s_{23}\, s_{45}} - \cancel{\frac{1}{s_{34}\, s_{12}}} + \frac{1}{s_{45}\, s_{12}} - \frac{1}{s_{12}\, s_{23}}$$

$$\frac{1}{s_{23}\, s_{45}} + \frac{1}{s_{45}\, s_{12}} - \frac{1}{s_{12}\, s_{23}}$$

$$1 + \frac{s_{23}}{s_{12}} - \frac{s_{45}}{s_{12}} \Rightarrow \frac{\boxed{s_{12} + s_{23} - s_{45}}}{s_{23}\, s_{12}\, s_{45}} = 0$$

$$p_2 \cdot (p_1 + p_3 + p_4 \cdot p_5) = 0$$

$$s_{12} + \cancel{s_{23}} + \cancel{s_{24}} + \cancel{s_{25}} = 0$$

$$p_i \left( \sum_j p_j \right) = 0 \Rightarrow \sum_j s_{ij} = 0$$

$$\begin{aligned}
i = 2 \quad & s_{21} + s_{23} + \cancel{s_{24}} + \cancel{s_{25}} = 0 \quad \oplus \\
i = 4 \quad & \cancel{s_{42}} + \cancel{s_{43}} + s_{45} = 0 \quad \ominus \\
i = 3 \quad & s_{32} + \cancel{s_{34}} + \cancel{s_{35}} = 0 \quad \oplus \\
i = 5 \quad & -s_{12} + \cancel{s_{52}} + \cancel{s_{53}} + s_{54} = 0 \quad \ominus
\end{aligned}$$

$$s_{12} + s_{23} - s_{45}$$

$$2 \left( s_{21} + s_{23} - s_{45} \right) = 0$$

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$$s_{13} = s_{14} = 0$$

$$s_{24} = s_{25} = 0$$